Amirhesam Taherzadegan

Photogrammetrist , Computer vision developer

Taherzadehesam@gmail.com   HesamTaherzadeh   HesamTaherzadeh

Portfolio (hesamtaherzadeh.github.io)

H.Taherzadegan

Professional Experience

2023-09 – present
Suzhou, China

SLAM Algorithm Engineer / Core Engineer
Software Motion

- **Led HDMap and OfflineMap development for the JAC Automotive project**, provided centimeter-level precision on 5 different scenarios
- **Optimized lane line tracking and localization**, improving self-driving accuracy by designing robust code with GPS disruption handling for ADAS.
- **Developed Loop Closure SDK,** a multi-modal loop closure module, both LiDAR and visual.
- **Developed an IMU pre-integration module**, boosting sensor fusion accuracy in the mapping system.
- **Mentored 5 and managed 3 members in total,** designed onboarding procedure, attended **5** technical interviews, and co-designed technical written test

2021-09 – 2024-01
Tehran, Iran

Head of Mathematics for AI Teaching Assistant Team (Graduate course)
Khajeh Nasir University of Technology

- Co-designed **course project**
- **Supervised** students' and other TAs' mathematical correctness.

Course website

2021-09 – present
Tehran, Iran

Head of least squares optimizations and hypothesis testing Teaching Assistant (TA) Team
Khajeh Nasir University of Technology

- Held **+50 hours** of class and online content for the course
- **Taught** the practical part of the course (**Practical Convex Optimization**).

Youtube playlist

2022-06 – 2022-09
Tehran, Iran

Embedded computer vision developer
Devspec

Developed a deep learning-based Lane Departure Warning system for **ARM CPUs**, reaching **6 FPS** on RPI and **15 FPS** on Jetson Nano

2022-02 – 2022-07
Tehran, Iran

Computer vision Developer
Khajeh Nasir University of Technology

- Developed a full-fledge desktop application for 3D reconstruction.
- Implemented **multi-threading** to process over **50** full-size images with OpenGL and QT

Education

2023-09 – present
Tehran, Iran

Master of Science in Photogrammetry
Khajeh Nasir University of Technology

GPA: 19.3/20 (4/4)
Photogrammetry is the process of performing accurate measurements on images, using 3D computer vision
Thesis: Deep Dense, Metric, Monocular VI-SLAM

2019-09 – 2023-06
Tehran, Iran

Bachelor of Science in Geomatics engineering
Khajeh Nasir University of Technology

GPA : 18.45/20 (4/4)
Thesis : Developing a deep learning based visual odometry system for ARM CPUs, First Rank of Class of 2019 in geomatics at KNTU

Awards

Master's scholarship
National Organization of Educational Testing
Accepted in the Photogrammetry MSc. program at KNTU through the Iran National Elites Foundation (INEF) scholarship

RobotX fellowship
ETH Zurich
Accepted into a fellowship program in ETH Zurich for the RobotX center with a 6% acceptance rate - Hosting lab: Computer vision and Geometry

Outstanding student
KNTU
Recognized as the outstanding master's student in geomatics engineering, achieving the highest GPA across four majors .

Research interest

- Visual/LiDAR/Inertial SLAM
- 3D Scene understanding
- NeRFs, and Gaussian Splatting (Novel View synthesis)
- 3D reconstruction

Publications

Dense Metric Monocular SLAM based on RTAB-Map
ISPRS Geo spatial Conference 2025
submitted

Dense Metric-Accurate Visual-Inertial SLAM based on ORB-SLAM3
Under Development

Skills

C++ Programming
OOP, Multi-threading, Design pattern, Memory management

Robotics Perception
Traditional & learned approaches, Trackings, and 3D scene understanding
- *PCL, OpenCV, Open3D*

ROS 1/2
Extensive experience with Melodic, Noetic, Humble - RViZ, Gazeboo

Probabilistic Robotic and Computer vision
Pose Graph Optimization
- GTSAM, Ceres Solver

Python Programming
OOP, Design patterns and Scientific computations

SLAM
Filter-based & PGO-based *LiDAR , Visual, Visual-inertial*

Deep learning Model & Engineering
- PyTorch, TensorFlow, ONNX and Quantizations

Sensor Fusion and Calibration
Kalman Filter family, IMU-Image intrinsics, IMU-LiDAR-Camera Calibs
- *Kalibr, Allan*

Languages

Persian
Native

English
- **IELTS 8.0 (EXPIRY 2027)**

Projects

Indoor Monocular Depth Estimation
Supervised approach
Monocular depth estimation network for Indoor environment trained on DIODE(Dense Indoor and Outdoor DEpth) dataset

Deep learning based LDW system
written in C++ and python and also has been tested in Jetson Nano 2GB and Raspberrty pi 3b and has 3 different outputs regarding the warning

Meta-heuristic Regression
Utilizing GA to avoid overfitting of parameters in C++

PCST
Geo-localization of satelite imagery with UI in Qt (C++/Python)

Visual odometry and Triangulation using OpenCV C++ API
Self localization(6 DOF) and Triangulation of feature points of a calibrated camera embedded in raspberry pi 3b, GPS-enabled using an external module

Bundle Adjustment from Scratch
Python based program that will accept two image coordinates and GCPs(Ground Control Points) and will perform bundle adjustment to adjust the GCPs and triangulate tie points

Target-less monocular BEV image creator
Used deep learning methods to regress a homography matrix to create bird-eye view (ROS-enabled)

3D monocular object detection
Utilized YOLO to detect cars for autonomous driving scenarios through deep-learning based monocular depth estimation (ROS-enabled)

Courses

Deep Learning Specialization (5 course)
Deep learning.AI

Introduction to Statistics
Stanford Online

Tensorflow developer and Advanced Tensorflow developer (8 courses)
Deep learning.AI